

Expt-8: Unknown Capacitance by Schering Bridge

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1 Aim of the Experiment

To find the value of the unknown capacitance and its reactance with the help of the Schering bridge.

2 Theory Used

The Schering bridge is an AC bridge used to measure an unknown capacitance and its dielectric loss. It compares an imperfect capacitor C_X with a standard loss-free capacitor. At bridge balance, the detector gives a null indication, and the impedance balance condition is satisfied. By separating the real and imaginary parts of the balance equation, the unknown capacitance and equivalent loss resistance can be determined. The capacitive reactance is then calculated using the measured value of C_X .

3 Formula Used

From the Schering bridge balance condition, separating the real and imaginary parts gives:

$$R_X = R_2 \frac{C_1}{C_2}$$

and

$$C_X = C_2 \frac{R_1}{R_2}$$

where:

C_X = unknown capacitance,

R_X = equivalent loss/series resistance of the unknown capacitor,

R_1 = resistance adjusted using the potentiometer,

R_2 = selected resistance,

C_1 = standard capacitor,

C_2 = fixed capacitor.

4 Observations

Value of Capacitor-1 (C_1) = 0.001 μF

Value of Capacitor-2 (C_2) = 0.47 μF

Table 1: Observation Table for C_{X_1}

Sr. No.	R_1 (k Ω)	R_2 (k Ω)	Calculated Capacitance C_{X_1} (μF)	Calculated Resistance R_{X_1} (k Ω)
1	1.5	2.8	0.251	0.00595
2	2	3.7	0.254	0.00787
3	2.5	4.73	0.248	0.01006
4	3	5.72	0.246	0.01217
5	3.5	6.54	0.251	0.01391
6	4	7.6	0.247	0.01617

Table 2: Observation Table for C_{X_2}

Sr. No.	R_1 (k Ω)	R_2 (k Ω)	Calculated Capacitance C_{X_2} (μF)	Calculated Resistance R_{X_2} (k Ω)
1	1.5	2.2	0.320	0.00468
2	2	2.9	0.324	0.00617
3	2.5	3.6	0.326	0.00766
4	3	4.3	0.328	0.00914
5	3.5	5.05	0.326	0.01074
6	4	5.75	0.327	0.01223

Table 3: Observation Table for C_{X_3}

Sr. No.	R_1 (k Ω)	R_2 (k Ω)	Calculated Capacitance C_{X_1} (μF)	Calculated Resistance R_{X_1} (k Ω)
1	1.5	1.55	0.455	0.00329
2	2	2.05	0.458	0.00436
3	2.5	2.6	0.452	0.00553
4	3	3.1	0.455	0.00659
5	3.5	3.6	0.57	0.00766
6	4	4.1	0.458	0.00872

5 Analysis on observations

For C_{X_1} ,

Slope (m) = 1.9148571428571424

Intercept (b) = -0.08419047619047462

C_2 = 0.47

C_{X_1} = 0.24544911966577146

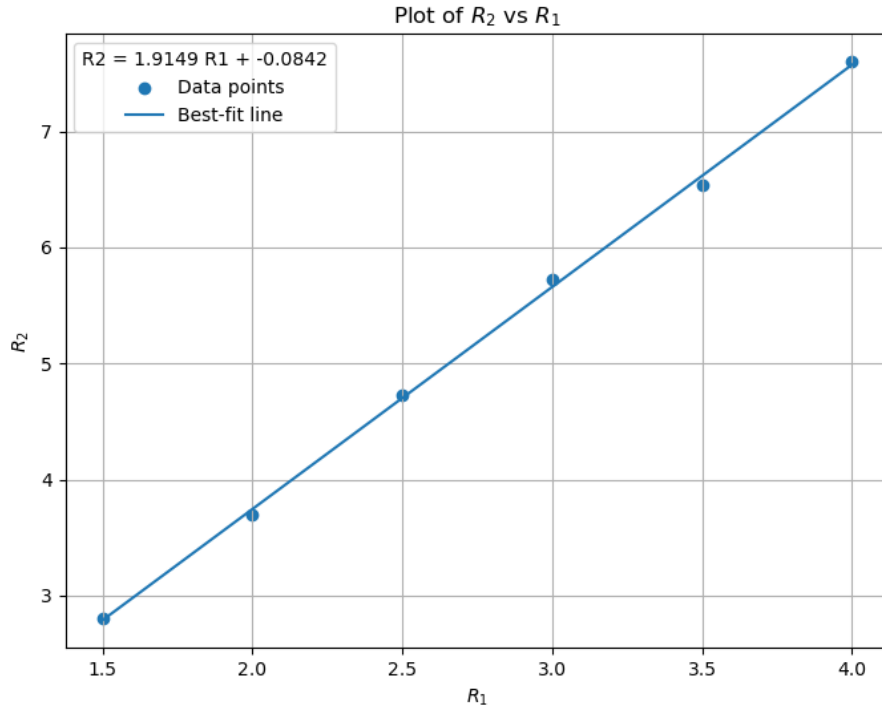


Figure 1: R_2 vs R_1 for C_{X_1}

For C_{X_2} ,

Slope (m) = 1.4228571428571424

Intercept (b) = 0.05380952380952507

$C_2 = 0.47$

$C_{X_2} = 0.33032128514056236$

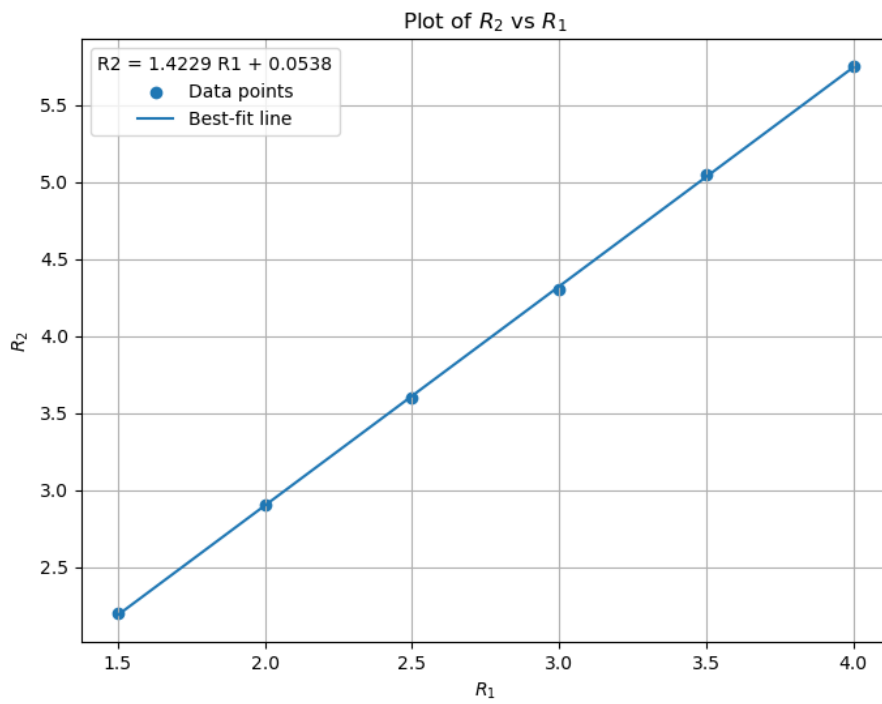


Figure 2: R_2 vs R_1 for C_{X_2}

For C_{X_3} ,

Slope (m) = 1.0228571428571427

Intercept (b) = 0.020476190476191102

$C_2 = 0.47$

$C_{X_3} = 0.45949720670391064$

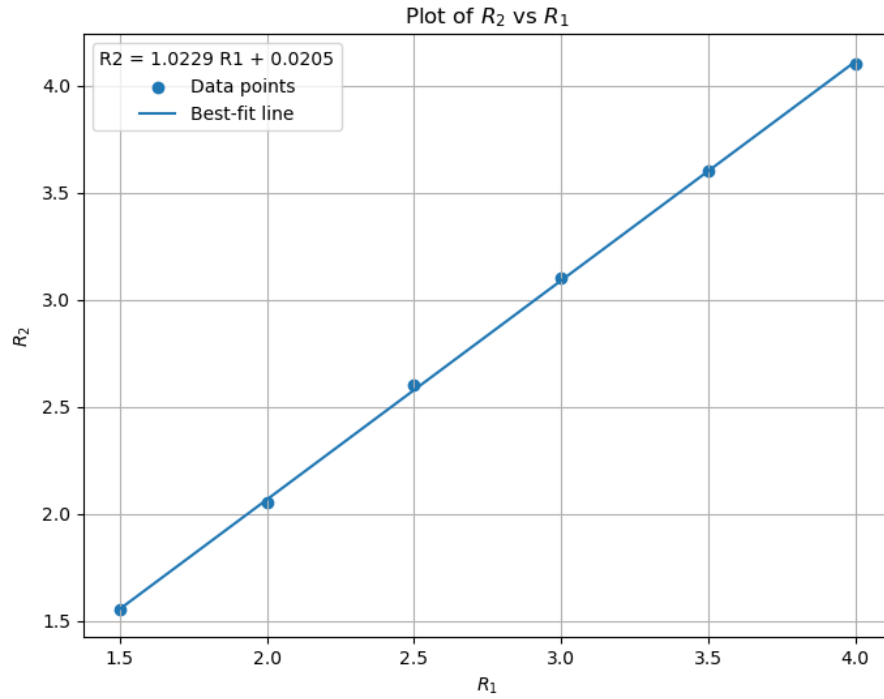


Figure 3: R_2 vs R_1 for C_{X_3}

6 Error analysis

For C_{X_1} ,

Slope (m) = 1.914857 ± 0.027431

$C_2 = 0.47$

$C_{X_1} = 0.245449 \pm 0.003516$

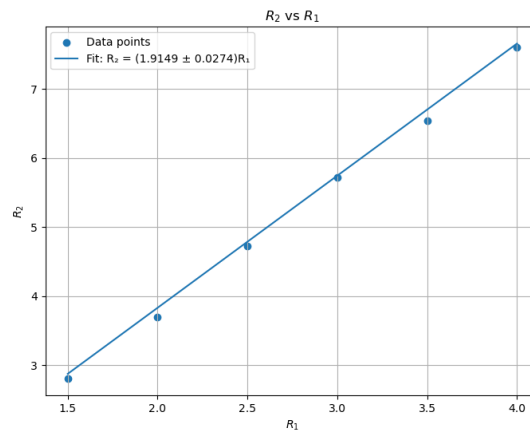


Figure 4: R_2 vs R_1 for C_{X_1}

For C_{X_2} ,

Slope (m) = 1.422857 ± 0.007737

$C_2 = 0.47$

$C_{X_1} = 0.330321 \pm 0.001796$

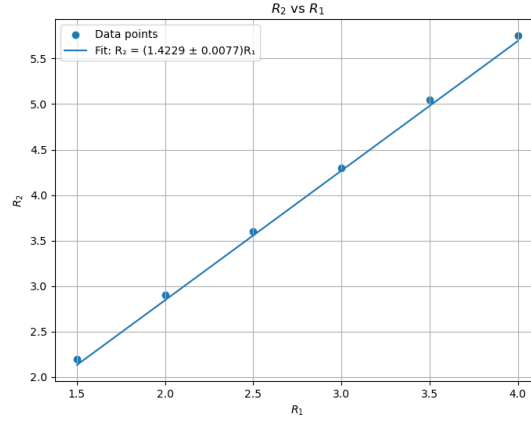


Figure 5: R_2 vs R_1 for C_{X_2}

For C_{X_3} ,

Slope (m) = 1.022857 ± 0.007737

$C_2 = 0.47$

$C_{X_3} = 0.459497 \pm 0.003476$

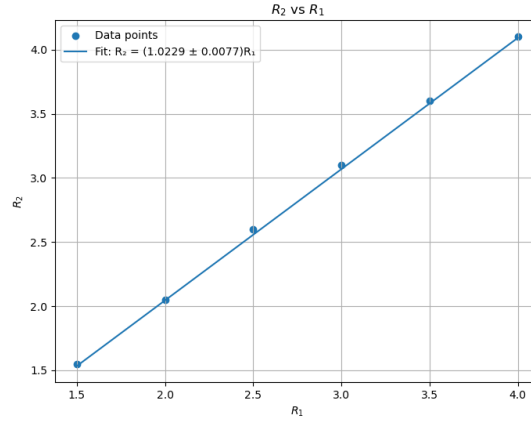


Figure 6: R_2 vs R_1 for C_{X_3}